

# Final Review

Tuesday April 28th

Final May 6th 1-3pm

## Virtual Memory

- Separation of user logical memory from physical memory
  - only part of the problem needs to be in main memory
  - logical address space can be much larger
  - allows address spaces to be shared by several processes
  - allows for more efficient process creation
- Go to the GitHub link in the slides to see the demonstration

## Page Fault

- The memory you need to access is not loaded in the page table
- OS will look at another table to decide:
  - invalid reference -> abort
  - just not in memory -> page-in
- Get an empty frame
- Swap (read) page into the new frame
- set validation bit = 1
- Restart instruction

## Page Replacement

- Find the location of the desired page on disk
- Find a free frame
  - if there is a free frame use it
  - if there is no free frame, use a page
- replacement algorithm to select a victim frame
- Read the desired page into the newly free frame and update the page and frame tables.
- Restart the process

## First-in-first-out FIFO Algorithm

- simple queue
- watch class demo

## Optimal Algorithm

- You know the sequence
- Replace the page that will not be used for the longest time in the future
- Watch class demo
- Provable optimal has the lowest fault rate
- used as a yardstick to compare other algorithms but not practical as you need to know the full sequence to use

## Least Recently Used

- Replace the page that has not been used for the longest amount of time in the past (**oldest**)
- Focused on past information instead of future
- watch the class demo

## Least Frequently Used

- keep track of the frequency